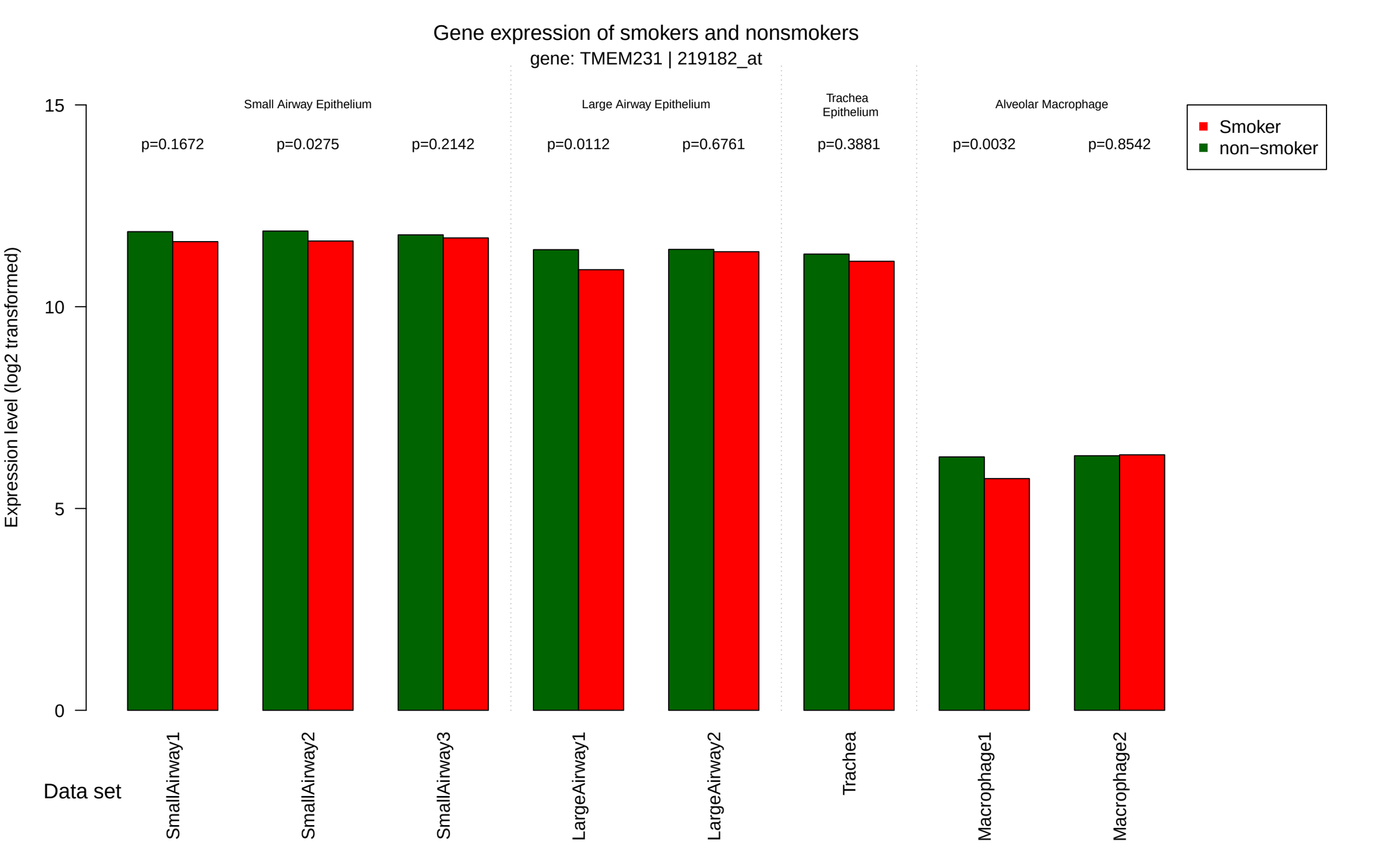
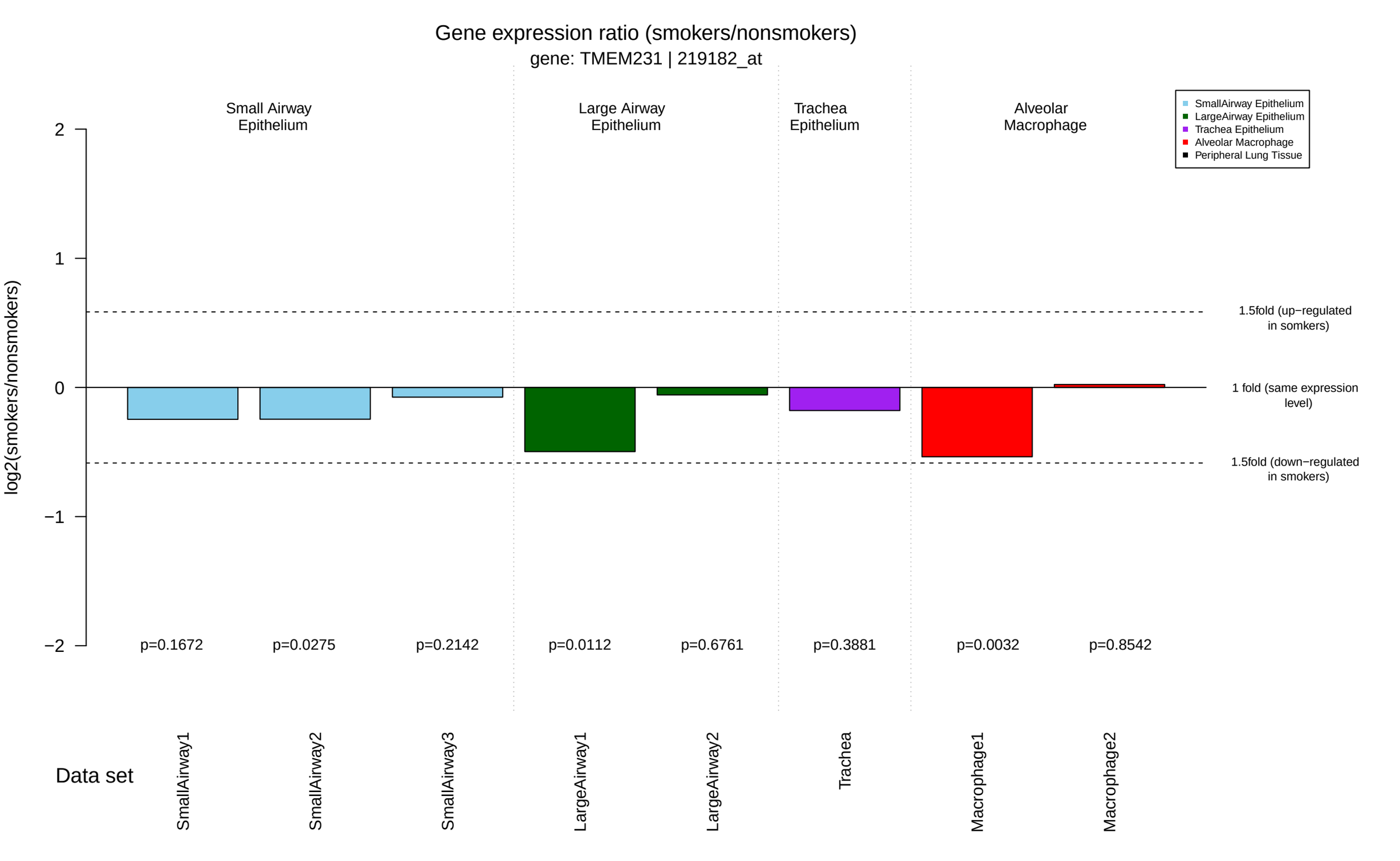
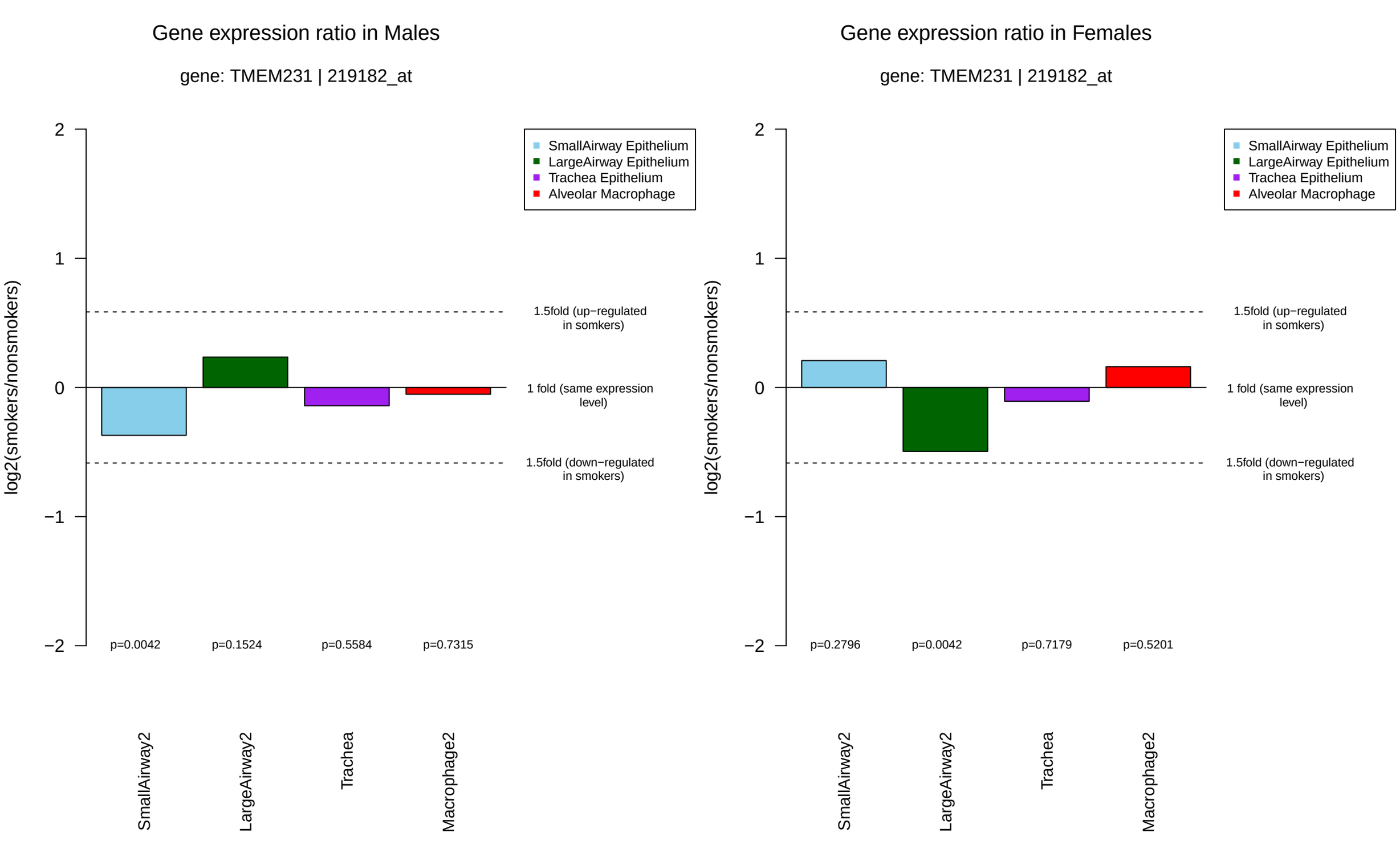
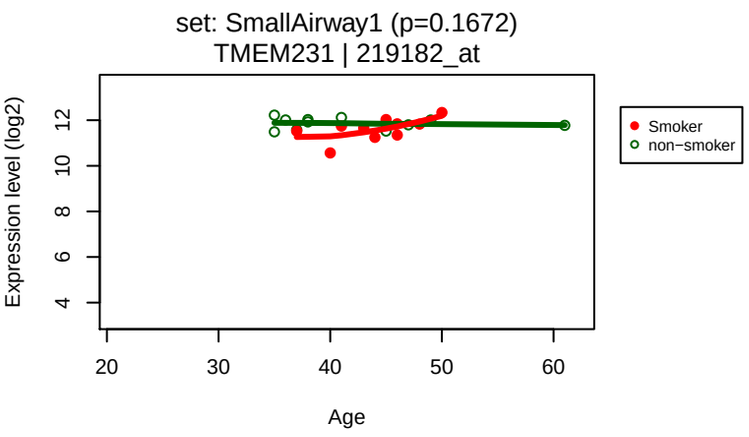
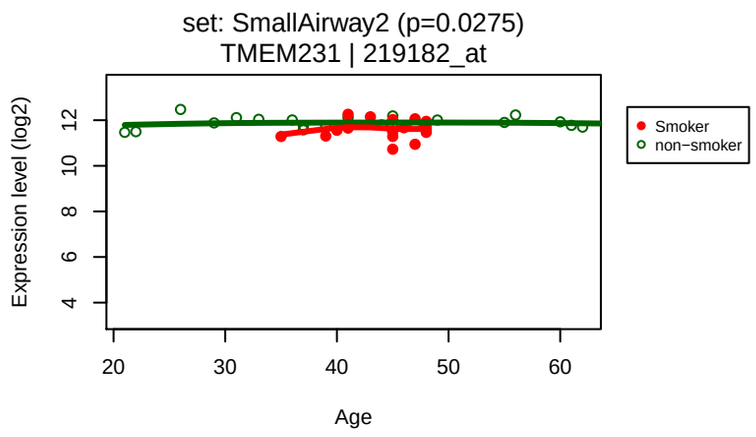
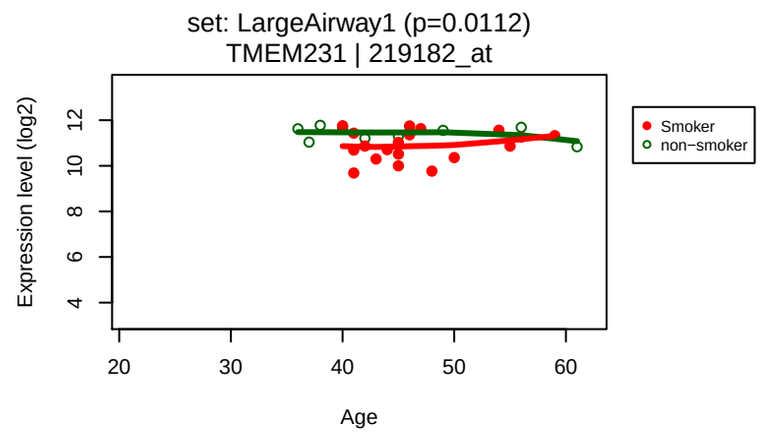

Gene expression of smokers and nonsmokers
gene: TMEM231 | 219182_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1672
p=0.0275
p=0.2142
p=0.0112
p=0.6761
p=0.3881
p=0.0032
p=0.8542
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

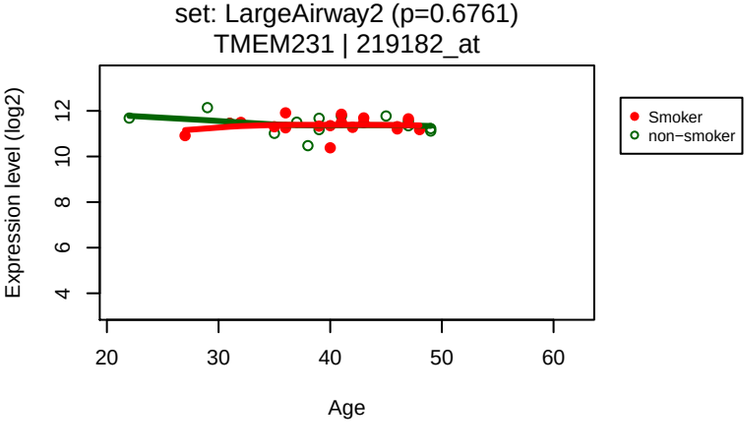

Gene expression ratio (smokers/nonsmokers)
gene: TMEM231 | 219182_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1672
p=0.0275
p=0.2142
p=0.0112
p=0.6761
p=0.3881
p=0.0032
p=0.8542
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

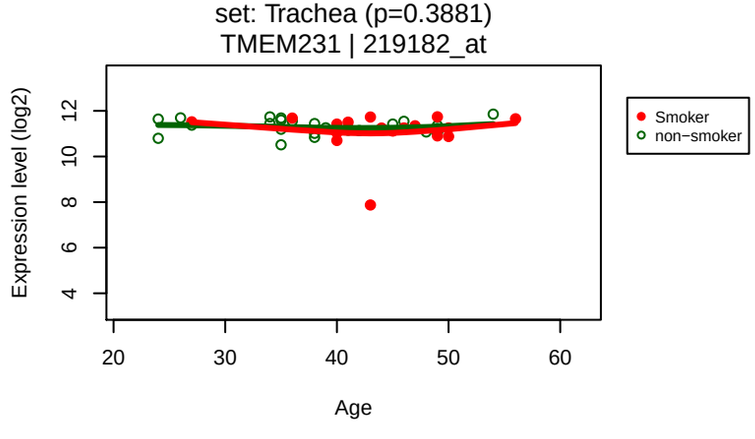

Gene expression ratio in Males
gene: TMEM231 | 219182_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0042
p=0.1524
p=0.5584
p=0.7315
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM231 | 219182_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2796
p=0.0042
p=0.7179
p=0.5201
SmallAirway2
LargeAirway2
Trachea
Macrophage2

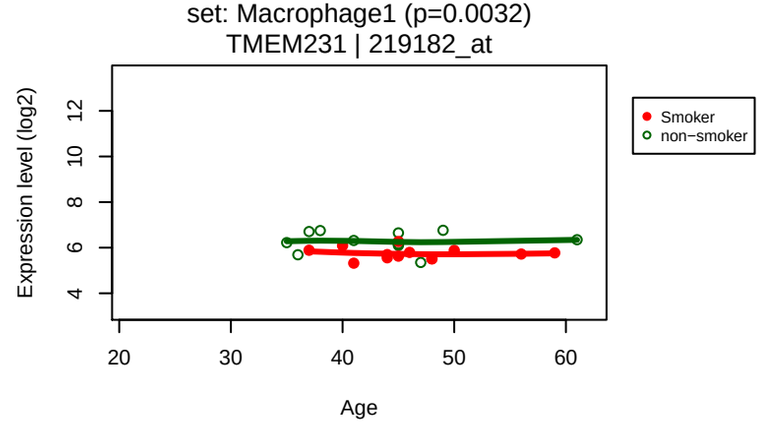

set: SmallAirway1 (p=0.1672)
TMEM231 | 219182_at
Expression level (log2)
Age
Smoker
non−smoker

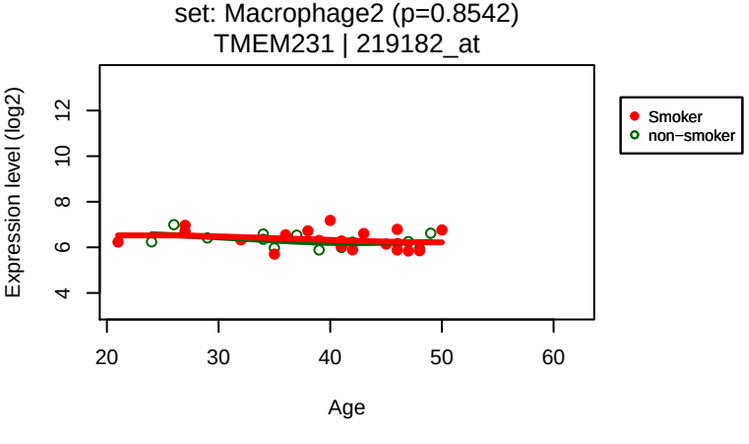

set: SmallAirway2 (p=0.0275)
TMEM231 | 219182_at
Expression level (log2)
Age
Smoker
non−smoker

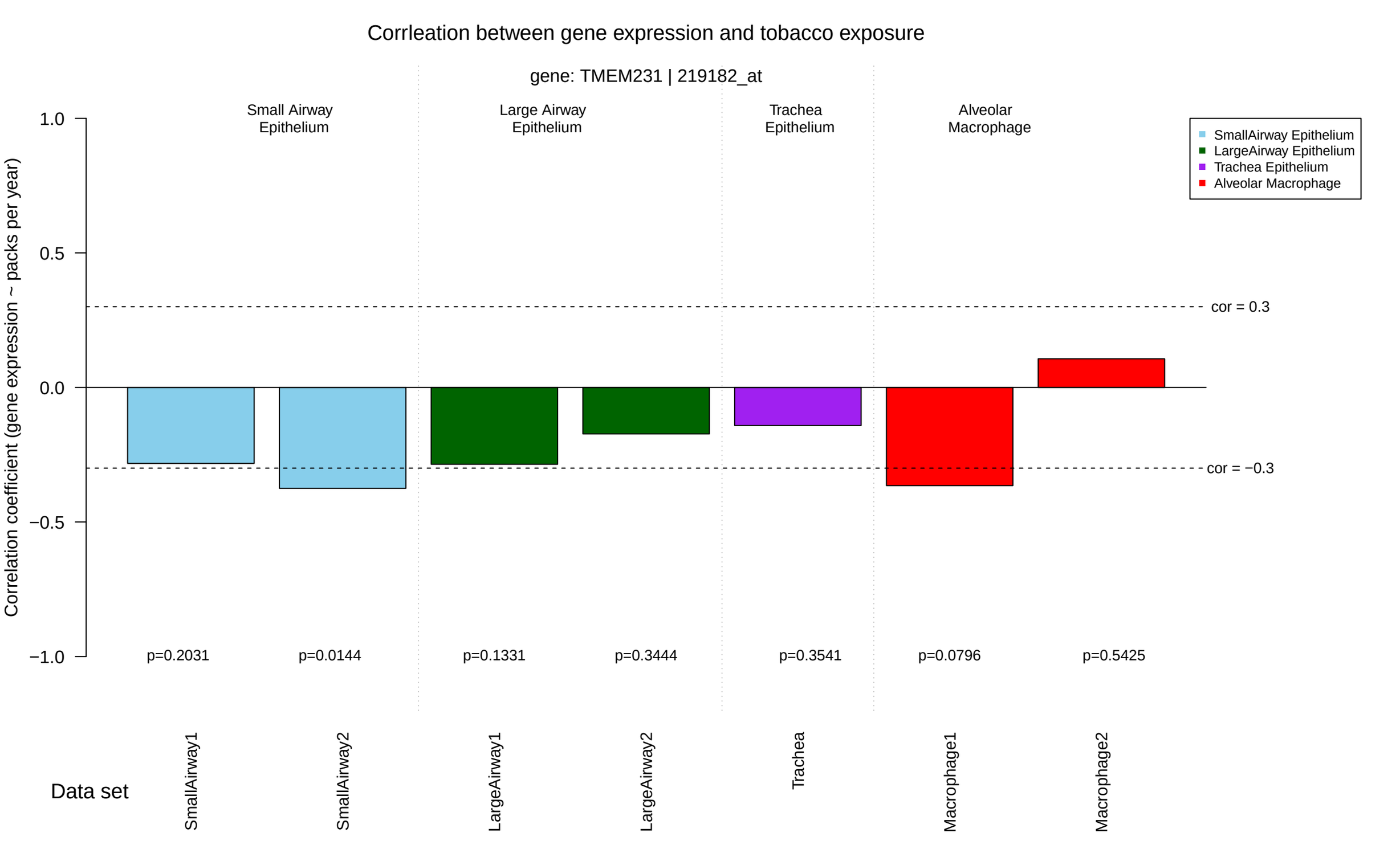

set: LargeAirway1 (p=0.0112)
TMEM231 | 219182_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6761)
TMEM231 | 219182_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3881)
TMEM231 | 219182_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0032)
TMEM231 | 219182_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8542)
TMEM231 | 219182_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM231 | 219182_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2031
p=0.0144
p=0.1331
p=0.3444
p=0.3541
p=0.0796
p=0.5425
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2